

The [Agricultural Competitiveness White Paper](#), released in 2015, included a large number of initiatives and commitments by the Australian Government to strengthen Australia's agriculture sector that were delivered over 4 years.

The White Paper was a \$4 billion investment in Australian farmers and the competitiveness and profitability of the sector, including \$200 million for biosecurity surveillance and analysis to protect Australia's animal and plant health status. This provided additional resources for four broad themes: strengthening surveillance, building community-based engagement, growing scientific capability and information and analysis.

A number of animal health-related projects were funded, including: further development of a data sharing and coordination software system for pest and disease surveillance (AUSPestCheck); antimicrobial resistance projects; foot-and-mouth disease training; national avian influenza wild bird surveillance; expansion of proficiency testing stocks for foot-and-mouth disease; and development of national testing capability for Classical Swine Fever.

Funding for AUSPestCheck facilitated the provision and sharing of animal pest and disease surveillance data as a Proof of Concept and enabled the collating of information to support the National Animal Health Information System reporting. AUSPestCheck is a data sharing and coordination software system for pest and disease surveillance managed by Plant Health Australia (PHA). The Proof of Concept system was developed by PHA through a Commonwealth-funded National Landcare Program Innovation grant in 2014 and subsequently improved and enhanced through other State and Commonwealth contributions including under the Agricultural Competitiveness White Paper.

Other projects focused on antimicrobial resistance (AMR). Those supported antimicrobial resistance surveillance in the Australian chicken egg industry to estimate the susceptibility profiles against specified antimicrobials in indicator bacteria; a surveillance project in the chicken meat industry which resulted in the National Biosecurity Manual for Chicken Growers being updated to include the potential for transfer of antimicrobial resistant bacteria from humans to chickens. The project for AMR surveillance in the Australian barramundi industry allowed the industry to obtain baseline data on the AMR status of farmed Australian barramundi – this data will deliver evidence to support market access and/or inform the need to modify industry practices if required. A collaborative project with University of Sydney was funded to extend the Veterinary Antimicrobial Stewardship Online Training Program, an educational tool to assist veterinarians to reduce and rationalise their use of antimicrobial agents in clinical practice and address the spread of antimicrobial resistance.

Continued investment in national surveillance for influenza viruses in wild birds supported diagnostic test capability, laboratory capacity, and field sampling capability, and allowed inferences to be made about the source of infection when influenza viruses are detected in poultry and pigs. The project contributed to meeting national and international commitments relating to trade and reporting of avian influenza to the World Organisation for Animal Health. Communication of information produced by this project will assist in maintaining partnerships between governments (Commonwealth, State and Territory) and industry, including to strengthen messaging to enhance producer awareness of biosecurity risks. Also, analysis of multiple years of wild bird surveillance data facilitates more efficient

surveillance planning and sampling frames for future surveillance, as well as identifying risk factors for the presence of virus subtypes in Australian wild bird populations.

White Paper funding supported private Australian veterinarians' participation in real time foot-and-mouth disease training in Nepal. This training provided the veterinarians with the skill set necessary to identify and if necessary manage an outbreak of foot-and-mouth disease.

White Paper funding also supported laboratory diagnostics. The projects funded in this area included the validation of diagnostic tests for Ostreid Herpesvirus-1 Microvariant; Validation of Diagnostic tests for White Spot Syndrome Virus; New Assays and Enhanced Quality Assurance within the Laboratories for Emergency Animal Disease Diagnosis and Response Network; Quality Assured National Capacity for Sequencing and Comparative Analysis of Genomes for Animal Disease Surveillance; and Diagnosis and Enhancing the Laboratories for Emergency Animal Disease Diagnosis and Response Network's Capability on Animal Influenza Serology. The validation details from the Ostreid herpesvirus-1 (OsHV-1) project will be used to design a surveillance program for OsHV-1 by the Australia-wide Pacific oyster industry and relevant jurisdictions, and were included in the World Organisation for Animal Health's diagnostic manual. The validation of diagnostics tests for white spot syndrome virus (WSSV) details have been used to design the national surveillance program for WSSV in wild crustaceans, and will also be included in the World Organisation for Animal Health's diagnostic manual.

A project titled New Assays to Enhance Quality Assurance within the Laboratories for Emergency Animal Disease Diagnosis and Response (LEADDR) Network enabled serological capacity for classical swine fever and foot-and-mouth disease to be deployed as needed. This capacity has also been quality-assured through the established network's quality assurance programs, including its proficiency testing program and internal quality assurance activities.

Laboratory diagnostics-related work also included a project on national capacity for sequencing and comparative analysis of genomes for animal disease surveillance diagnosis. This project brought about a minimum standard for the use of next generation sequencing (NGS) based on diagnostic capability in emergency animal disease including national guidelines for quality assurance of implementation of a national diagnostic NGS capacity in relevant laboratories. This work will contribute to the setting of international standards in this area.

Source: <https://www.agriculture.gov.au/agriculture-land/animal/health/animal-health-related-projects>